

Shapes & Measures



SYLLABUS 3.1 3.2 3.3 3.4

You cannot pace out the Earth. So how did anyone find its size? Not by walking. The answer came from something much smaller: an angle.

Check the fact yourself. On flat ground the horizon is level with your eye. Climb a hill and it drops slightly below your eye line, and the higher you climb, the further it drops. That drop is the **dip**, and it is small: about half a degree from a good hill, the width of the Moon in the sky.

Your line of sight does not cut into the Earth. It touches the surface and passes on, meeting the radius at that point at a right angle. So the hilltop, the horizon and the centre of the Earth form one right-angled triangle, with hill height h and dip θ :

$$\cos \theta = \frac{R}{R+h}, \quad \text{so} \quad R = \frac{h \cos \theta}{1 - \cos \theta}.$$

Measure the hill, measure the dip, and the radius comes out close to the modern 6371 km, from one hill, in one afternoon. Al-Bīrūnī did exactly this a thousand years ago, from the fort of Nandana in what is now Pakistan.

Size decides more than shape. Double an object and its surface grows fourfold while its volume grows eightfold, so bigger things carry less surface for what they hold. That is why crushed ice melts fast, and why no insect could be scaled up to the size of a horse.

By the end of this chapter you will find the angle between a line and a plane in a solid, work out the volume and surface area of standard solids, solve any triangle with the sine and cosine rules, ambiguous case included, handle elevation, depression and bearings, and measure in **radians**, which turn arc length and sector area into one multiplication each.

A calculator is assumed throughout this chapter. The **EXACT** badge marks the questions whose answer must be left exact.

7.1 Three-Dimensional Geometry

The measurements you need from a solid are often not the ones marked on its faces. The diagonal of a room, or the angle a roof strut makes with the floor: neither of these lies along a face. Both run through the inside of the object, and coordinates are how you reach them.

Before any measuring, a word on names. Examination papers use a fixed notation for points, segments, lines and angles, and they use it without explanation.

Notation	Meaning
$A(x, y)$	the point A with coordinates x and y
$[AB]$	the line segment with endpoints A and B
AB	the length of $[AB]$
(AB)	the line through A and B , continuing both ways
$\hat{ABC}$	the angle at B , between $[BA]$ and $[BC]$
$\triangle ABC$	the triangle with vertices A , B and C

The middle letter is the one to hold on to: it names the vertex. So $\hat{ABC}$ and $\hat{CBA}$ are the same angle, while $\hat{BAC}$ is a different one. Note too that the length of a segment carries no bars. Bars are reserved for the magnitude of a vector, which Chapter 9 introduces.

Coordinates in the plane extend to space by adding a third axis, z . The distance formula extends with them. In two dimensions it is Pythagoras' theorem. In three dimensions it is Pythagoras used *twice*: once across the base, then once upward.

KEY · IB The distance between (x_1, y_1, z_1) and (x_2, y_2, z_2) is

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$

The midpoint is the average of the coordinates, $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2}\right)$.

Pythagoras is used twice. Across the base it gives

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

Treat that as one side of a new right triangle. Its other side is the vertical rise, $(z_2 - z_1)$, and

Pythagoras on this second triangle gives the formula.

The equation of a sphere comes from the same formula. A sphere is the set of points at a fixed distance r from a centre (a, b, c) , so its equation is $(x - a)^2 + (y - b)^2 + (z - c)^2 = r^2$.

7.1.1 The angle between a line and a plane

A line that is not parallel to a plane meets it at an angle. Many angles can be measured at that meeting point, so one of them has to be chosen, and mathematics chooses the smallest. Picture a lamp directly above the line, shining straight down. The line casts a shadow on the plane, and the angle between the line and its own shadow is the smallest angle available. That shadow is called the **projection** of the line onto the plane.

DEF The **angle between a line and a plane** is the angle between the line and its projection on the plane. To find it, drop a perpendicular from a point on the line down to the plane. The line, its projection and that perpendicular form a right triangle, and right-angled trigonometry solves it.

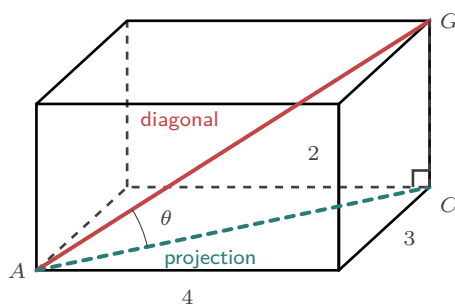


Figure 7.1 The space diagonal AG and its shadow AC on the base. The vertical edge CG meets the base at a right angle, so ACG is a right triangle and θ is the angle between the diagonal and the base.

EXAMPLE 7.1

A cuboid has a 4×3 base and height 2. Find the angle between the space diagonal AG and the base.

The diagonal of the base is $AC = \sqrt{4^2 + 3^2} = 5$. This is the projection of AG onto the base, and $CG = 2$ is vertical, meeting the base at a right angle. In the right triangle ACG ,

$$\tan \theta = \frac{CG}{AC} = \frac{2}{5} \implies \theta = \arctan(0.4) \approx 21.8^\circ.$$

The order is the same each time. Find the projection on the plane, then the perpendicular height, then use $\tan$.

7.1.2 The angle between two intersecting lines

Two lines that meet also make an angle, and inside a solid those two lines are usually edges from the same corner. Take the two edges, then join their far ends. The three segments

form a triangle, and the angle you want sits at the corner. Measure the three sides with the distance formula, then solve the triangle.

EXAMPLE 7.2

A square-based pyramid has base corners $A(0, 0, 0)$, $B(6, 0, 0)$, $C(6, 6, 0)$, $D(0, 6, 0)$ and apex $V(3, 3, 4)$. Find the angle between the slant edges VA and VB .

The two edges meet at V and span the triangle VAB , so the angle we want is $\widehat{AVB}$. First measure the triangle. By the distance formula,

$$VA = \sqrt{(3-0)^2 + (3-0)^2 + (4-0)^2} = \sqrt{34},$$

and $VB = \sqrt{34}$ by the same calculation, while $AB = 6$ along the base.

The triangle is isosceles, so cut it in half. The midpoint of AB is $M(3, 0, 0)$, and VM meets AB at a right angle. In the right triangle VMA , the side $AM = 3$ is opposite the half-angle at V , and the hypotenuse is $VA = \sqrt{34}$:

$$\sin \frac{\theta}{2} = \frac{3}{\sqrt{34}} \Rightarrow \frac{\theta}{2} \approx 31.0^\circ \Rightarrow \theta \approx 61.9^\circ.$$

The two formulas of this section carry the whole solution: the distance formula for the three sides, the midpoint to cut the triangle in half, then right-angled trigonometry for the angle.

YOUR TURN 7.1 Three-Dimensional Geometry

1. Find the distance between each pair of points.

(a) $(1, 2, 3)$ and $(4, 6, 3)$

(b) $(0, 0, 0)$ and $(2, 3, 6)$

(c) $(-1, 0, 2)$ and $(2, -4, 2)$

(d) $(2, -1, 4)$ and $(5, 3, 4)$

2. Answer each part.

(a) Find the midpoint of $(1, 2, 3)$ and $(5, 4, 1)$ (b) Find the midpoint of $(2, -1, 4)$ and $(6, 3, 0)$.

3. Write the equation of each sphere.

(a) centre $(0, 0, 0)$, radius 5

(b) centre $(1, 2, 3)$, radius 4

(c) centre $(2, -1, 3)$, radius 6

(d) centre $(0, 0, 2)$, radius $\sqrt{7}$

4. A cuboid has base 3×4 and height 12.

(a) Find the length of the diagonal of the base.

(b) Find the length of the space diagonal.

(c) Find the angle between the space diagonal and the base.

7.2 Volumes and Surface Areas of Solids

You already know the cuboid, the prism and the cylinder. Three curved solids appear beside them again and again. What matters is where each formula comes from, because a formula you can rebuild is a formula you will not misapply.

	Sphere, radius r	$V = \frac{4}{3}\pi r^3, \quad A = 4\pi r^2$
KEY · IB	Cone, radius r , height h	$V = \frac{1}{3}\pi r^2 h, \quad \text{curved } A = \pi r l$
	Pyramid, base area B , height h	$V = \frac{1}{3}Bh$
	Cylinder, radius r , height h	$V = \pi r^2 h, \quad \text{curved } A = 2\pi r h$

The cylinder is listed in the *Prior learning* table at the front of the booklet rather than beside the other three.

Each formula has a short reason behind it.

- **Cylinder.** Cut the curved side along one vertical line and unroll it into a rectangle of height h and width $2\pi r$, the base circumference. Its area is $2\pi r h$.
- **Cone.** The same cut unrolls the curved side into a sector. Its radius is the **slant height** l , measured from the rim to the tip, and its arc is the base circumference $2\pi r$. A sector's area is half its arc times its radius, so the curved area is $\frac{1}{2}(2\pi r)(l) = \pi r l$.
- **The $\frac{1}{3}$.** A cube cuts into three identical square pyramids. Each one stands on a face of the cube, and all three tips meet at the opposite corner. Each is therefore a third of the cube, and the same fraction holds for every pyramid and every cone: $V = \frac{1}{3}Bh$.
- **Sphere.** Put a sphere inside the smallest cylinder that holds it, of radius r and height $2r$. Archimedes proved the sphere is two thirds of that cylinder. The cylinder holds $\pi r^2(2r) = 2\pi r^3$, so the sphere holds $\frac{4}{3}\pi r^3$. Its surface, $4\pi r^2$, is the area of four **great circles** (a circle round the sphere at its widest).

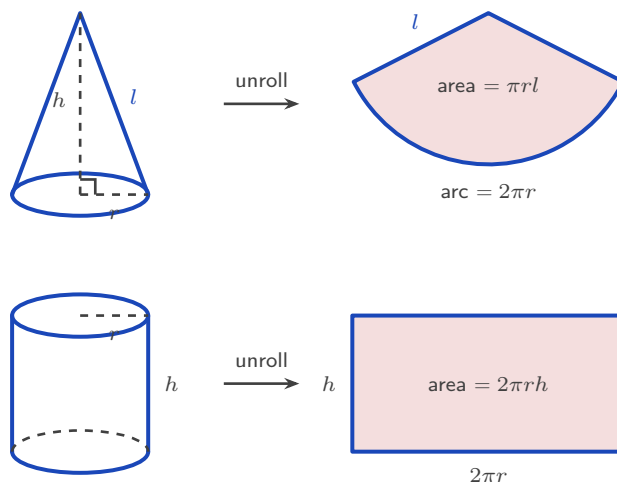


Figure 7.2 Cut the curved surface of a cone along one slant line and it opens into a sector of radius l whose arc is the base circumference. Cut a cylinder the same way and it opens into a rectangle. Each curved-area formula is the area of the flat shape it opens into.

Two mistakes are common. The cone's curved surface uses the slant height l , not the vertical height h ; Pythagoras links them by $l = \sqrt{r^2 + h^2}$. And "total surface area" means every face: a solid hemisphere has a curved part $2\pi r^2$ and a flat base πr^2 , giving $3\pi r^2$, while the same hemisphere as an open bowl has only $2\pi r^2$.

EXAMPLE 7.3

A solid is a cone of base radius 8 and height 6 sitting on top of a hemisphere of radius 8. Find its total surface area.

The exposed surface is the cone's curved part plus the hemisphere's curved part. The two flat circles where they join are hidden inside, so neither counts. The cone has slant height

$$l = \sqrt{8^2 + 6^2} = 10,$$

so its curved area is $\pi r l = 80\pi$. The hemisphere's curved area is $2\pi r^2 = 128\pi$, and the total is

$$80\pi + 128\pi = 208\pi \approx 653.$$

Deciding which faces are exposed is the harder step; the formulas themselves are routine.

7.2.1 Similar solids

Every formula above is a constant times a length squared, for an area, or a length cubed, for a volume. That fixes what happens when a solid is enlarged.

Two solids are **similar** when they have the same shape and all corresponding lengths are in the same ratio. That ratio is the **linear scale factor** k . Multiplying every length by k multiplies $4\pi r^2$ by k^2 and $\frac{4}{3}\pi r^3$ by k^3 . Every formula here behaves the same way, because each depends on length raised to one fixed power.

KEY **Scale factors.** If two similar figures have lengths in the ratio k , then

$$\frac{\text{length}_2}{\text{length}_1} = k, \quad \frac{\text{area}_2}{\text{area}_1} = k^2, \quad \frac{\text{volume}_2}{\text{volume}_1} = k^3.$$

One ratio of lengths settles every area and every volume.

A cone cut parallel to its base is the case to recognise. The small cone above the cut has the same apex and the same slope, so it is similar to the whole, and one ratio of heights gives its radius, its surface and its volume at once.

EXAMPLE 7.4

A cone has base radius 9 cm and height 12 cm. A cut parallel to the base is made 4 cm below the apex. Find the radius of the circular cut and the volume of the small cone above it.

The small cone is similar to the whole cone, and the two heights are known, so

$$k = \frac{4}{12} = \frac{1}{3}.$$

The radius is a length, so it scales by k :

$$r = \frac{1}{3}(9) = 3 \text{ cm}.$$

Similar triangles say the same thing directly: the vertical axis and the radius of the cut form a triangle similar to the one made by the axis and the base radius, so $\frac{r}{4} = \frac{9}{12}$.

The whole cone holds

$$V = \frac{1}{3}\pi(9)^2(12) = 324\pi \text{ cm}^3,$$

and volume scales by k^3 , so the small cone holds

$$\left(\frac{1}{3}\right)^3 (324\pi) = \frac{1}{27}(324\pi) = 12\pi \text{ cm}^3.$$

Computing it directly gives $\frac{1}{3}\pi(3)^2(4) = 12\pi$, which agrees. The piece left below the cut is not similar to either cone, so no scale factor applies to it; its volume is the difference, $324\pi - 12\pi = 312\pi \text{ cm}^3$.

YOUR TURN 7.2 Volumes and Surface Areas of Solids

5. EXACT Find the volume of each solid. Leave your answers in terms of π where appropriate.

- | | |
|---|-------------------------------------|
| (a) sphere of radius 3 | (b) cone of radius 3, height 4 |
| (c) pyramid with base area 20, height 9 | (d) cylinder of radius 4, height 10 |

6. EXACT Find the surface area asked for in each case.

- | | |
|--|--|
| (a) the surface area of a sphere of radius 5 | (b) the curved surface area of a cone of radius 5, slant height 13 |
| (c) the total surface area of a solid hemisphere of radius 4 | (d) the curved surface area of a cylinder of radius 4, height 10 |

7. EXACT A cone has radius 6 and height 8.

- (a) Find its slant height.
 (b) Hence find its curved surface area.

8. **EXACT** A solid is a cone of radius 3 and height 4 standing on a hemisphere of radius 3.

- Explain which surfaces are exposed.
- Find the total surface area, in terms of π .
- Find the total volume, in terms of π .

7.3 The Sine and Cosine Rules

Right-angled triangles are solved with $\sin$, $\cos$ and $\tan$. Name the sides opposite, adjacent and hypotenuse relative to an angle, and $\sin \theta = \frac{o}{h}$, $\cos \theta = \frac{a}{h}$, $\tan \theta = \frac{o}{a}$. Most triangles have no right angle. Two rules extend trigonometry to all of them, and between them they solve any triangle from three known parts.

Both rules use one labelling. A side takes the lower-case letter of the angle *opposite* it, so side a faces angle A .

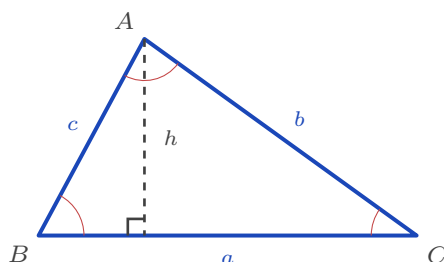


Figure 7.3 Each side is named after the angle opposite it: side a faces angle A , and so on. The dashed altitude h splits the triangle into two right triangles, which is where the sine rule comes from.

7.3.1 The sine rule

Drop a perpendicular of height h from A onto side a , as in the figure above. It cuts the triangle into two right triangles, and in each one h is the side opposite a known angle. On the left, $h = c \sin B$. On the right, $h = b \sin C$. Both expressions measure the same segment, so $c \sin B = b \sin C$, which rearranges to $\frac{b}{\sin B} = \frac{c}{\sin C}$. Dropping a perpendicular from a different corner gives the third ratio.

KEY · IB **Sine rule.** In any triangle,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

The sine rule pairs a side with the angle facing it. Use it whenever you know one complete pair, plus one more side or angle. The standard shorthand writes S for a known side and

A for a known angle, in the order they occur round the triangle. These are the cases AAS, ASA and SSA.

7.3.2 The ambiguous case

The sine rule needs one extra check when you use it to find an *angle*.

Suppose you know two sides and an angle that is *not* between them. This is the SSA case. The sine rule gives an equation of the form $\sin B = k$, and between 0° and 180° that equation has two solutions: the angle B and its supplement $180^\circ - B$. An angle and its supplement have the same sine, so the equation cannot separate them.

Your calculator returns only the first. Test the second yourself. Add it to the angle you were given: if the total is still below 180° , a positive third angle remains, so a second triangle exists and both answers are correct.

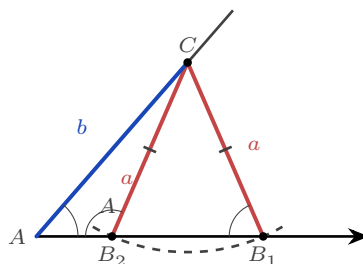


Figure 7.4 Two sides and a non-included angle. Swinging the fixed length a from C meets the base at B_2 and at B_1 , so two different triangles share the same a , b and A . The angles at B_2 and B_1 are supplementary — here 113.5° and 66.5° — which is why the sine rule returns both.

EXAMPLE 7.5

In triangle ABC , $a = 9$, $b = 11$, and $A = 40^\circ$. Find angle B .

By the sine rule, $\frac{\sin B}{11} = \frac{\sin 40^\circ}{9}$, so

$$\sin B = \frac{11 \sin 40^\circ}{9} \approx 0.786.$$

The calculator gives $B \approx 51.8^\circ$. The supplement is $180^\circ - 51.8^\circ = 128.2^\circ$, so test it: $A + B = 40^\circ + 128.2^\circ = 168.2^\circ$, which is below 180° . A third angle of 11.8° remains, so that triangle exists too.

Both values of B are correct. There are two triangles, and a complete answer reports both.

CAUTION This check is needed only when the sine rule gives an *angle*. If you are finding a *side*, or using the cosine rule, no supplement arises. The cosine rule returns an angle between 0° and 180° directly, because $\cos$ is negative for obtuse angles and positive for acute ones, so it distinguishes the two.

7.3.3 The cosine rule

When the known angle sits *between* the two known sides, no side is paired with the angle opposite it, so the sine rule cannot start. The cosine rule covers this case. Read it as Pythagoras' theorem with a correction term for the angle not being 90° .

Coordinates prove it directly. Put the corner C at the origin with side a along the x -axis, so B sits at $(a, 0)$. The point A lies a distance b from the origin, in the direction C . By the definitions of cosine and sine, its coordinates are $(b \cos C, b \sin C)$. The distance formula now measures the third side:

$$c^2 = (a - b \cos C)^2 + (b \sin C)^2 = a^2 - 2ab \cos C + b^2 \cos^2 C + b^2 \sin^2 C,$$

and $\cos^2 C + \sin^2 C = 1$, so the last two terms add to b^2 .

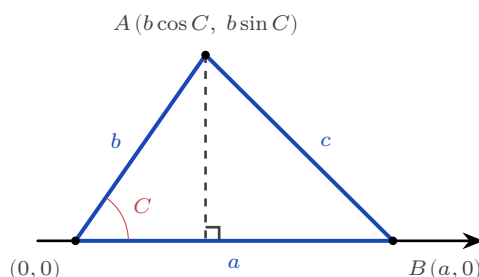


Figure 7.5 Placing C at the origin with a along the axis fixes all three corners. Only A needs trigonometry, and the distance from A to B is then the cosine rule.

KEY · IB **Cosine rule.** In any triangle,

$$c^2 = a^2 + b^2 - 2ab \cos C, \quad \text{equivalently} \quad \cos C = \frac{a^2 + b^2 - c^2}{2ab}.$$

The letters rotate. Written about a different corner the same rule reads $a^2 = b^2 + c^2 - 2bc \cos A$, and the pattern is the same in each version: the side opposite the named angle stands alone on the left.

When $C = 90^\circ$, $\cos C = 0$ and the rule becomes $c^2 = a^2 + b^2$, which is Pythagoras' theorem. The first form finds a third side from two sides and the angle between them (SAS). The rearranged form finds an angle from all three sides (SSS).

7.3.4 The area of a triangle

The familiar area $\frac{1}{2} \text{ base} \times \text{height}$ becomes a trigonometric formula as soon as the height is written in terms of a side. Take b as the base. The perpendicular height from B down to it is the opposite side of the right triangle at C , so $h = a \sin C$ and

$$\text{Area} = \frac{1}{2} b (a \sin C).$$

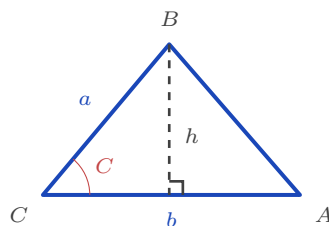


Figure 7.6 The height of the triangle on base b is the opposite side of the right triangle at C , so it is $a \sin C$. Substituting it into $\frac{1}{2} \times \text{base} \times \text{height}$ gives the formula.

KEY · IB **Area of a triangle.** $\text{Area} = \frac{1}{2} ab \sin C$, using two sides and the angle between them.

This formula needs exactly what the cosine rule needs, so one sketch usually answers both questions at once.

EXAMPLE 7.6

A triangular field has $AB = 90$ m, $AC = 70$ m, and the angle at A is 110° . Find the third side and the area of the field.

The known angle lies between the two known sides, so the cosine rule finds BC . Rotate the letters to put A in the named position:

$$BC^2 = 90^2 + 70^2 - 2(90)(70) \cos 110^\circ \approx 17\,309 \implies BC \approx 132 \text{ m.}$$

The area needs the same three pieces of information, so no further measurement is required:

$$\text{Area} = \frac{1}{2}(90)(70) \sin 110^\circ \approx 2960 \text{ m}^2.$$

Two sides and the angle between them fix a triangle completely. Once you hold that combination, every other part of the triangle can be found.

What you are given	What to use
Two angles and any side (AAS, ASA)	sine rule
Two sides and an angle opposite one (SSA)	sine rule, plus the ambiguous case
Two sides and the angle between them (SAS)	cosine rule, for the third side
All three sides (SSS)	cosine rule rearranged, for an angle
The same (SAS), when the area is wanted	$\text{area} = \frac{1}{2} ab \sin C$

Read the table by counting what you know. Three parts of a triangle determine the rest, and the arrangement of those three parts is what chooses the rule.

YOUR TURN 7.3 The Sine and Cosine Rules

9. Use the sine rule to find the required side or angle.

- (a) $a = 7$, $A = 40^\circ$, $B = 60^\circ$: find b (b) $a = 9$, $A = 35^\circ$, $B = 55^\circ$: find b
(c) $A = 45^\circ$, $B = 60^\circ$, $a = 10$: find C and b (d) $a = 10$, $b = 12$, $A = 35^\circ$: find the acute angle B

10. Use the cosine rule.

- (a) $b = 5$, $c = 8$, $A = 60^\circ$: find a (b) $a = 6$, $b = 9$, $C = 110^\circ$: find c
(c) $a = 6$, $b = 7$, $c = 8$: find angle A (d) sides 5, 6, 7: find the largest angle

11. Find the area of each triangle.

- (a) sides 5 and 8, included angle 30° (b) $a = 9$, $b = 6$, $C = 90^\circ$
(c) sides 7 and 9, included angle 40°

12. In triangle ABC , $a = 7$, $b = 9$ and $A = 45^\circ$.

- (a) Find the two possible values of angle B .
(b) Show that both give a valid triangle.
(c) Explain why this situation cannot arise when the given angle is included between the two given sides.

7.4 Elevation, Depression and Bearings

Real questions do not arrive as labelled triangles. They arrive as heights, distances and directions: how far away is the boat, what heading takes the ship home. Two conventions turn that language into a triangle you can solve.

An **angle of elevation** is measured upward from the horizontal to a line of sight. An **angle of depression** is measured downward from it. The observer's horizontal and the ground are parallel lines. So the angle of depression from a cliff top down to a boat equals the angle of elevation from the boat up to the cliff top. They are alternate angles, and either one can be used in the triangle.

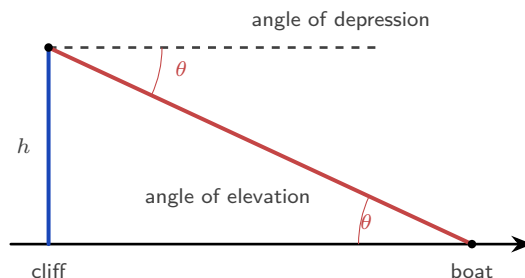


Figure 7.7 The dashed line at the top is the observer's horizontal, and the ground is parallel to it. The line of sight crosses both, so the two marked angles are alternate angles and are equal.

One sighting is never enough to find a height, because the distance to the object is unknown as well: two unknowns, one equation. A second sighting from a known distance nearer supplies the missing equation. The height is the same from both positions, so writing it twice and equating the two expressions eliminates the distance.

EXAMPLE 7.7

From a point on level ground the angle of elevation of the top of a tower is 25° . From a point 30 m nearer the tower, on the same straight line, the angle of elevation is 40° . Find the height of the tower.

Let h be the height and let d be the distance from the *nearer* point to the foot of the tower. The far point is then $d+30$ from the foot. Each sighting gives a right triangle with h opposite the angle and the ground distance adjacent to it, so

$$h = d \tan 40^\circ \quad \text{and} \quad h = (d + 30) \tan 25^\circ.$$

Both measure the same tower, so set them equal and collect the terms in d :

$$d \tan 40^\circ = (d + 30) \tan 25^\circ \implies d (\tan 40^\circ - \tan 25^\circ) = 30 \tan 25^\circ,$$

$$d = \frac{30 \tan 25^\circ}{\tan 40^\circ - \tan 25^\circ} \approx \frac{13.99}{0.3728} \approx 37.5 \text{ m.}$$

The height follows from either equation:

$$h = 37.5 \tan 40^\circ \approx 31.5 \text{ m.}$$

Check it against the other sighting: the far point is 67.5 m away, and $\arctan\left(\frac{31.5}{67.5}\right) \approx 25^\circ$, as given. Measuring d from the nearer point is what keeps the second distance as $d+30$; taking d from the far point instead makes it $d-30$, and mixing the two is the usual source of an impossible answer.

A **bearing** gives a direction as an angle measured clockwise from north. It runs from 000° to 360° and is always written with three figures, so due east is 090° and not 90° . Three figures keep the reading unambiguous on a chart. Due south is 180° , due west is 270° , and north-west is 315° .

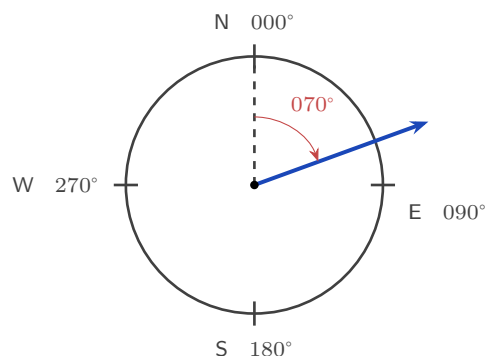


Figure 7.8 A bearing is measured clockwise from north, always with three figures. The arrow here is on a bearing of 070°.

The **back bearing** is the direction of the return journey, and it differs from the outward bearing by 180°. Add 180° when the bearing is below 180°, and subtract 180° when it is 180° or more. Either way the answer stays between 000° and 360°.

EXAMPLE 7.8

A ship sails 60 km from port P on a bearing of 050° to a point Q , then 80 km on a bearing of 140° to R . Find the distance PR .

The two bearings differ by $140^\circ - 50^\circ = 90^\circ$, so the ship turns through a right angle at Q : the angle PQR is 90° . Then PR is a hypotenuse, and Pythagoras gives

$$PR = \sqrt{60^2 + 80^2} = \sqrt{10000} = 100 \text{ km.}$$

Sketching the bearings from a north line at P and at Q is what shows the right angle. The arithmetic then follows from the picture.

Most journeys do not turn so conveniently. When the corner angle is not 90° , read the two distances and that angle off the same sketch. Find the missing distance with the cosine rule, then find the direction back with the sine rule. Most bearings questions have this shape, so the order of the steps is always the same.

KEY Solving a bearings problem.

1. Sketch the journey, and draw a north line at **every** point where a bearing is given.
2. Mark each bearing as an angle clockwise from its own north line.
3. Find the interior angle of the triangle at the corner. Take the back bearing of the first leg, then subtract the second bearing.
4. Use the cosine rule for the missing distance, then the sine rule for the missing angle.
5. Turn that angle back into a bearing by adding it to the bearing of the first leg.

Students often leave out step 1. Without a north line drawn at the corner, step 3 cannot be done at all.

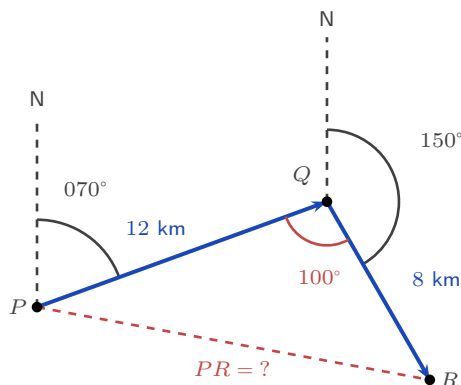


Figure 7.9 The two bearings are measured clockwise from a north line at P and at Q . Together they fix the interior angle $P\hat{Q}R = 100^\circ$, which is what the cosine rule needs.

EXAMPLE 7.9

A hiker walks 12 km from P on a bearing of 070° to Q , then 8 km on a bearing of 150° to R . Find the distance PR and the bearing of R from P .

First construct the angle at the corner. The bearing back from Q to P is $070^\circ + 180^\circ = 250^\circ$, and the onward bearing is 150° , so the interior angle is $P\hat{Q}R = 250^\circ - 150^\circ = 100^\circ$. Subtracting the second bearing from the back bearing is the standard method, and the sketch agrees: the angle at the corner is obtuse.

No right angle, so the cosine rule finds the closing side:

$$PR^2 = 12^2 + 8^2 - 2(12)(8) \cos 100^\circ \approx 241.3 \Rightarrow PR \approx 15.5 \text{ km.}$$

For the bearing of R from P , find the angle $Q\hat{P}R$ by the sine rule:

$$\frac{\sin(Q\hat{P}R)}{8} = \frac{\sin 100^\circ}{15.54} \Rightarrow \sin(Q\hat{P}R) \approx 0.507 \Rightarrow Q\hat{P}R \approx 30.5^\circ.$$

The bearing of R from P is therefore $070^\circ + 30.5^\circ \approx 100.5^\circ$, reported as 100° to the nearest degree. Every step came from the labelled diagram: the bearings give the corner angle, the cosine rule the closing side, and the sine rule the final direction.

YOUR TURN 7.4 Elevation, Depression and Bearings

13. Find the angle of elevation in each case.

- (a) a 50 m tower, from 120 m away on level ground (b) a 45 m tower, from 90 m away on level ground

14. In each case, find the horizontal distance from the base of the tall object to the object below.

- (a) From the top of an 80 m cliff, the angle of depression to a boat is 25° .
- (b) From the top of a 30 m building, the angle of depression to a car is 40° .
- (c) From the top of a 60 m cliff, the angle of depression to a buoy is 20° .

15. Answer each question about bearings.

- (a) Write the bearing of due west as a three-figure bearing.
- (b) A ship travels on a bearing of 120° . Find its back bearing.
- (c) B is on a bearing of 065° from A . Find the bearing of A from B .
- (d) Find the back bearing of 200° , and explain why you subtract here rather than add.

16. A ladder 6 m long rests against a vertical wall at 70° to the ground.

- (a) Find the height it reaches on the wall.
- (b) Find the distance from the foot of the ladder to the wall.

17. A ship sails 40 km from P on a bearing of 030° to Q , then 50 km on a bearing of 120° to R .

- (a) Show that angle $PQR = 90^\circ$.
- (b) Find the distance PR .
- (c) Find the bearing of R from P .

7.5 Radians, Arc Length, and Sector Area

The degree is a human invention. The Babylonians cut the circle into 360 parts, and we inherited the choice. Nothing about a circle requires 360. The circle does suggest a unit of its own: take the radius, bend it round the circumference, and measure the angle it makes at the centre. That angle is one **radian**. Because the circumference is $2\pi r$, exactly 2π radii fit around it, so a full turn is 2π radians and one radian is about 57.3° .

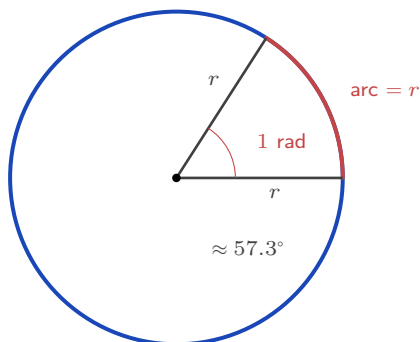


Figure 7.10 One radian is the angle at the centre for which the arc is exactly as long as the radius. A full turn takes 2π of them, because the circumference is 2π radii long.

KEY 2π radians = 360° , so π radians = 180° . Convert by multiplying: degrees to radians $\times \frac{\pi}{180}$; radians to degrees $\times \frac{180}{\pi}$.

The angles that appear most often should be known without conversion.

Degrees	30°	45°	60°	90°	180°	360°
Radians	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	2π

The reason for the new unit appears at once: arc length and sector area become simple fractions of the whole circle. An arc is the same fraction of the circumference as its angle θ is of a full turn, so $\frac{l}{2\pi r} = \frac{\theta}{2\pi}$, and the two 2π 's cancel.

KEY · IB For a sector of radius r and angle θ in radians,

$$\text{arc length } l = r\theta, \quad \text{sector area } A = \frac{1}{2}r^2\theta.$$

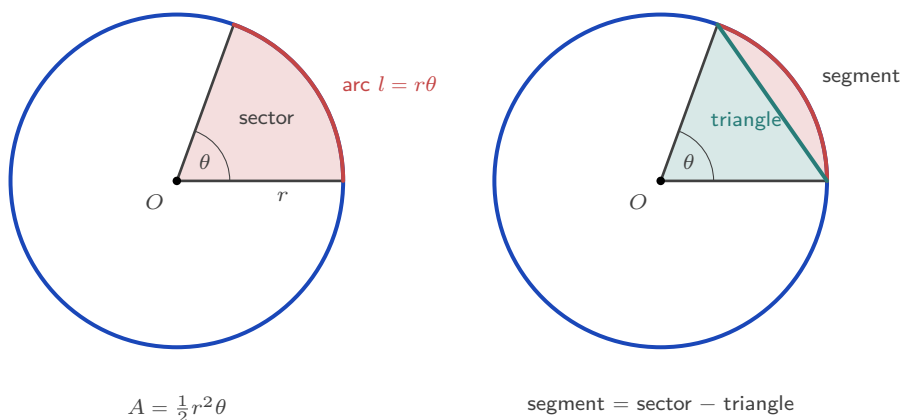


Figure 7.11 **Left:** the *sector* is the pie slice between two radii and the arc. **Right:** the chord cuts that same sector into a triangle and a *segment*, so subtracting the triangle's area from the sector's leaves the segment.

The area works the same way. The sector is the fraction $\frac{\theta}{2\pi}$ of the whole disc πr^2 , so $A = \frac{\theta}{2\pi} \cdot \pi r^2 = \frac{1}{2}r^2\theta$. A **segment** is the region between a chord and its arc. It is a sector with the triangle removed, and that triangle has two sides r with the angle θ between them, so §7.3 gives its area as $\frac{1}{2}r^2 \sin \theta$:

$$\text{segment} = \frac{1}{2}r^2\theta - \frac{1}{2}r^2 \sin \theta = \frac{1}{2}r^2(\theta - \sin \theta).$$

CAUTION Every formula in this section needs θ in **radians**. Substituting an angle in degrees gives an answer that is wrong by a factor of $\frac{180}{\pi}$, and the working looks correct, so the error is easy to miss. Check the mode on your calculator before you start.

EXAMPLE 7.10

A sector has radius 10 cm and angle $\frac{2\pi}{5}$ radians. Find its arc length and area.

Directly from the formulas,

$$l = r\theta = 10 \cdot \frac{2\pi}{5} = 4\pi \approx 12.6 \text{ cm}, \quad A = \frac{1}{2}r^2\theta = \frac{1}{2}(100)\frac{2\pi}{5} = 20\pi \approx 62.8 \text{ cm}^2.$$

For the segment cut off by the chord, subtract the triangle from the sector:

$$\frac{1}{2}r^2(\theta - \sin \theta) = 50 \left(\frac{2\pi}{5} - \sin \frac{2\pi}{5} \right) \approx 50(1.2566 - 0.9511) \approx 15.3 \text{ cm}^2.$$

Each formula took the angle in radians, exactly as given. The segment is far smaller than the sector, because most of the sector is triangle.

A sector's boundary is not the arc alone. Walk round the edge and you travel out along one radius, round the arc, and back along the other radius. The perimeter of a sector is therefore the arc plus two radii, and it has to be assembled from $l = r\theta$ rather than looked up, because the formula booklet does not print it.

KEY **Perimeter of a sector** of radius r and angle θ in radians:

$$P = 2r + r\theta = r(2 + \theta).$$

Two straight radii and one curved arc. This one is not in the formula booklet.

EXAMPLE 7.11

A sector has radius 7 cm and angle 1.5 radians. Find its perimeter.

The arc is

$$l = r\theta = 7(1.5) = 10.5 \text{ cm},$$

and the two radii add 14 cm, so

$$P = 2(7) + 10.5 = 24.5 \text{ cm}.$$

A quick sanity check: the two radii alone measure 14 cm, and an angle of 1.5 radians is under a quarter turn, so an arc a little longer than one radius is what the picture should show.

Given the perimeter and the area together, neither formula can be used on its own, because each contains both r and θ . Treat them as simultaneous equations, and eliminate θ .

EXAMPLE 7.12

A sector has perimeter 36 cm and area 45 cm^2 . Find its radius and its angle.

Write down both conditions:

$$2r + r\theta = 36, \quad \frac{1}{2}r^2\theta = 45.$$

The second gives $r^2\theta = 90$, so $\theta = \frac{90}{r^2}$. Substituting that into the first,

$$2r + r \cdot \frac{90}{r^2} = 36 \implies 2r + \frac{90}{r} = 36.$$

Multiply through by r and collect:

$$2r^2 - 36r + 90 = 0 \implies r^2 - 18r + 45 = 0 \implies (r - 3)(r - 15) = 0,$$

so $r = 3$ or $r = 15$. Test each against $\theta = \frac{90}{r^2}$. With $r = 3$ the angle is $\theta = 10$ radians, which is more than a full turn of $2\pi \approx 6.28$, so no such sector exists. With $r = 15$ the angle is $\theta = 0.4$ radians.

So $r = 15$ cm and $\theta = 0.4$ radians. Check: the arc is $15(0.4) = 6$ cm, giving a perimeter of $30 + 6 = 36$ cm, and the area is $\frac{1}{2}(225)(0.4) = 45 \text{ cm}^2$. ✓ The quadratic will usually offer two radii, and rejecting the one whose angle exceeds 2π is part of the answer, not an afterthought.

EXAMPLE 7.13

The opener measured the Earth from a hilltop. There is an older method, and it uses this section's formula. Eratosthenes measured the sun's rays at 7.2° to the vertical in Alexandria, while in Syene they were vertical. He took the arc between the towns to be 5000 stadia. Find the Earth's radius and circumference in stadia.

Convert the angle first:

$$7.2^\circ \times \frac{\pi}{180} = \frac{\pi}{25} \text{ radians.}$$

The two towns and the Earth's centre make a sector whose arc is the distance between them, so $l = r\theta$ can be used in reverse:

$$5000 = r \cdot \frac{\pi}{25} \implies r = \frac{125\,000}{\pi} \approx 39\,800 \text{ stadia,}$$

and the circumference is $2\pi r = 250\,000$ stadia, which is the one-fiftieth argument written as a formula. Taking a stadion as roughly 158 m gives a circumference near 39 500 km, within a couple of percent of the true 40 075 km. One angle, one arc, one formula, and the planet is measured.

YOUR TURN 7.5 Radians, Arc Length, and Sector Area

18. EXACT Convert to radians, in terms of π .

- (a) 60° (b) 135°
(c) 210° (d) 90°

19. Convert to degrees.

- (a) $\frac{3\pi}{4}$ (b) $\frac{2\pi}{3}$
(c) 1 radian, to three significant figures

20. Find the arc length and the area of each sector.

- (a) radius 8, angle 1.5 radians (b) radius 6, angle $\frac{\pi}{3}$ radians
(c) radius 9, angle 2.4 radians (d) radius 4, angle 90°

21. In each case the angle is unknown. Find it from the information given.

- (a) A sector has radius 10 and arc length 25. Find its angle in radians.
(b) A sector has radius 6 and area 54. Find its angle in radians, then find its arc length.

22. A chord subtends an angle of 2 radians at the centre of a circle of radius 8.

- (a) Find the area of the sector.
(b) Find the area of the triangle formed by the chord and the two radii.
(c) Find the area of the minor segment.
(d) Find the perimeter of the sector.

Chapter 7 · Shapes & Measures

Reflections

TOK The formulas $l = r\theta$ and $A = \frac{1}{2}r^2\theta$ are this simple only when θ is in radians, and later you will meet another case: the derivative of $\sin x$ is exactly $\cos x$, again only in radians. The degree is a human choice, made by the Babylonians. The radian is not a choice. It is the angle whose arc equals its radius, and the circle fixes it. Is the radian discovered, then, where the degree is invented?

IM Trigonometry was built by many people over many centuries. The Greek astronomer Hipparchus made the first table of chords around 150 BCE. In India, Aryabhata replaced the chord with the half-chord and called it the *jyā*, which is our sine. The word passed into Arabic, was misread there as *jaib*, meaning “bay”, and was translated into Latin as *sinus*. Islamic astronomers, al-Bīrūnī among them, extended the sine and cosine rules, partly to find the direction of Mecca from any city.

RW Fixing an unknown position from measured angles is called **triangulation**, and it is still how we map the world. Surveyors once carried chains of triangles across whole countries. Astronomers use *parallax*: as the Earth moves round the Sun a near star seems to shift against the far ones, and that small angle gives its distance. Each one is a triangle solved for a side nobody can reach.

RW Your phone does the same job with distances instead of angles. A GPS receiver times the signal from each satellite and turns each time into a distance, then finds the one point lying at all of those distances at once. Three satellites narrow it to a point on the Earth's surface, and a fourth corrects the receiver's clock. This is **trilateration**, and it is the sphere equation of §7.1 solved several times over.

EXT The ambiguous case is really a statement about congruence. Give a triangle by two angles and a side (ASA or AAS), by three sides (SSS), or by two sides and the angle between them (SAS), and only one triangle fits. Those are the congruence rules. Two sides and an angle that is *not* between them (SSA) is not one of them, so it can leave the triangle undecided. The sine rule's two answers are that gap showing through.

End-of-Chapter Problems — Chapter 7: Shapes & Measures

1. A cuboid has dimensions $6 \times 8 \times 10$.
- (a) Find the length of the space diagonal. [2]
(b) Find the angle between the space diagonal and the base. [3]
2. Consider the points $A(2, 0, 0)$, $B(0, 4, 0)$ and $C(0, 0, 4)$.
- (a) Find the lengths AB , AC and BC . [3]
(b) Find angle BAC . [3]
3. EXACT A cone has radius 5 and height 12.
- (a) Find the slant height. [1]
(b) Find the volume. [2]
(c) Find the total surface area. [2]
4. EXACT A solid consists of a cylinder of radius 3 and height 10 with a hemisphere of radius 3 on top. Find its total surface area. [5]
5. In triangle ABC , $a = 8$, $b = 11$, $c = 15$.
- (a) Find the largest angle. [3]
(b) Find the area of the triangle. [2]
6. In triangle PQR , $p = 10$, $q = 7$ and $R = 95^\circ$.
- (a) Find r . [2]
(b) Find the area of the triangle. [2]
7. A triangle has two sides of length 9 and 12 with an included angle of 40° .
- (a) Find the length of the third side. [2]
(b) Find the area of the triangle. [2]
8. From port P , a ship sails 40 km on a bearing of 030° to Q , then 50 km on a bearing of 120° to R .
- (a) Show that angle $PQR = 90^\circ$. [2]
(b) Find the distance PR . [2]

(c) Find the bearing of R from P . [3]

9. A sector has radius 12 cm and angle 1.2 radians.

(a) Find the arc length. [1]

(b) Find the area of the sector. [2]

(c) Find the perimeter of the sector. [2]

10. A sector has area 30 cm^2 and radius 6 cm.

(a) Find the angle of the sector in radians. [2]

(b) Find the arc length. [2]

11. A chord subtends an angle of 1.5 radians at the centre of a circle of radius 10 cm.

(a) Find the area of the sector. [2]

(b) Find the area of the triangle formed by the chord and the two radii. [2]

(c) Find the area of the minor segment. [2]

12. In triangle ABC , $A = 50^\circ$, $a = 6$ and $b = 7$.

(a) Find the two possible values of angle B . [3]

(b) Find the two possible areas of the triangle. [3]

13. EXACT A regular hexagon is inscribed in a circle of radius 8. Find its area. [4]

14. EXACT A cone has volume 100π and radius 5.

(a) Find the height. [2]

(b) Find the slant height. [2]

(c) Find the curved surface area. [1]

15. From a point on level ground, the angle of elevation of the top of a tower is 30° . Moving 40 m closer, the angle of elevation becomes 45° . Find the height of the tower. [5]

16. A square-based pyramid has base side 10 and height 12.

(a) Find the volume. [2]

(b) Find the slant height of a triangular face. [2]

(c) Find the total surface area. [2]

17. The area of a triangle is 24, with $a = 8$ and $b = 10$. Find the two possible values of the included angle C . [3]

18. Two ships leave a port at the same time. One sails on a bearing of 040° at 12 km/h, the other on a bearing of 100° at 16 km/h. Find the distance between them after 2 hours. [5]

19. Points $A(0, 0, 0)$ and $B(4, 4, 7)$ are given.

(a) Find AB . [2]

(b) Find the midpoint of $[AB]$. [1]

(c) Show that B lies on the sphere $x^2 + y^2 + z^2 = 81$. [1]

20. A pendulum of length 0.8 m swings through an angle of 0.5 radians. Find the length of the arc traced by the bob. [2]

21. EXACT Convert to radians, in terms of π .

(a) 30° [1] (b) 225° [1]

(c) 300° [1]

22. Two observers stand 200 m apart on level ground, with a balloon in the vertical plane between them. From one the angle of elevation to the balloon is 40° ; from the other it is 55° . Find the height of the balloon. [6]

23. In triangle ABC , $AB = 12$, $BC = 9$ and angle $BAC = 35^\circ$. Find the two possible values of angle BCA . [4]

24. Radians and arc length.

(a) Convert $\frac{5\pi}{6}$ to degrees. [1]

(b) Convert $\frac{7\pi}{4}$ to degrees. [1]

(c) Find the arc length of a sector of radius 9 and angle $\frac{5\pi}{6}$. [2]

25. A sector of a circle of radius 12 and angle 1.5 radians is rolled into a cone, so that the radius becomes the slant height and the arc becomes the base circumference. Find the radius and height of the cone. [6]

26. A sector of a circle has perimeter 20 and area 16. Find the radius r and angle θ . [6]

27. A chord of a circle of radius 10 has length 12. Find the area of the minor segment it cuts off. [6]

28. The area of a triangle is 84, and two of its sides have lengths 13 and 14. Find both possible lengths of the third side. [6]

29. A yacht leaves harbour H and sails 25 km on a bearing of 048° to a buoy A . It then sails 18 km on a bearing of 130° to a second buoy B .

- (a) Show that angle $HAB = 98^\circ$. [3]
- (b) Find the distance HB . [4]
- (c) Find the bearing of B from H . [4]
- (d) Find the area of triangle HAB . [3]
- (e) The yacht returns directly from B to H . Find the total distance it has sailed. [2]

30. A rectangular box $ABCDEFGH$ has a horizontal base $ABCD$ with $AB = 8$ cm and $BC = 6$ cm, and vertical edges of length 5 cm, so that G is the vertex above C .

- (a) Find the length of the base diagonal AC . [2]
- (b) Find the exact length of the space diagonal AG . [3]
- (c) Find the angle that AG makes with the base $ABCD$. [3]
- (d) Find the angle $G\hat{A}B$ between the diagonal AG and the edge AB . [4]
- (e) Find the total surface area of the box. [3]

31. EXACT A container is a cone of base radius 6 cm and height 15 cm, held with its point downwards.

- (a) Find the volume of the cone when full, in terms of π . [2]
- (b) Water is poured in to a depth of 10 cm. Using similar triangles, find the radius of the circular water surface. [3]
- (c) Find the volume of water, in terms of π . [3]
- (d) Find the fraction of the cone's volume that is filled, and explain why it is not $\frac{2}{3}$ even though the water reaches $\frac{2}{3}$ of the height. [4]
- (e) Find the depth of water when the cone is exactly half full, giving your answer to three significant figures. [3]

Challenge Questions

32. How many triangles fit the data? In triangle ABC you are given the side b , the angle A , and the side a opposite A . This is the SSA case. This investigation finds exactly how many triangles fit, for every possible value of a .

Throughout, take A to be acute, and let h be the distance from C to the line AB .

- (a) Show that $h = b \sin A$. [2]
- (b) Take $b = 10$ and $A = 40^\circ$. Find the number of triangles when $a = 5$, when $a = 8$, and when $a = 12$. [4]

- (c) Explain why no triangle exists when $a < b \sin A$. (*Hint: what is the shortest distance from C to the line AB ?*) [3]
- (d) Explain why exactly one triangle exists when $a = b \sin A$, and state what is special about it. [3]
- (e) Show that two triangles exist exactly when $b \sin A < a < b$, and explain what goes wrong at $a = b$. [4]
- (f) Now let A be obtuse. State how many triangles exist for $a \leq b$ and for $a > b$, and justify each answer. [4]

33. The best cone from a sector. A sector of radius R and angle θ radians is cut from a sheet of paper. The two straight edges are joined to make a cone. This investigation finds the angle that gives the largest cone.

Write $t = \frac{\theta}{2\pi}$, so that $0 < t < 1$.

- (a) The arc of the sector becomes the circumference of the base. Show that the base radius is $r = Rt$. [3]
- (b) Show that the height of the cone is $h = R\sqrt{1-t^2}$. [3]
- (c) Hence show that the volume is $V = \frac{1}{3}\pi R^3 t^2 \sqrt{1-t^2}$. [3]
- (d) Taking $R = 1$, use technology to find the value of t that makes V largest, and hence find θ to the nearest degree. [4]
- (e) Show that at that maximum $\frac{r}{R} = \sqrt{\frac{2}{3}}$, and find the exact value of $\frac{h}{r}$. [5]

34. How reliable was al-Bīrūnī's measurement? The opener described his method.

From a hill of height h he measured the angle of dip θ to the horizon, and the radius of the Earth follows from one right-angled triangle. This investigation asks how much trust the answer deserves.

- (a) The line of sight touches the Earth, so it meets the radius there at a right angle. Show that $\cos \theta = \frac{R}{R+h}$, and hence that $R = \frac{h \cos \theta}{1 - \cos \theta}$. [3]
- (b) Al-Bīrūnī recorded $h = 652$ cubits and $\theta = 34'$. Find R in cubits. (*Hint: $34' = \frac{34}{60}$ of a degree*) [3]
- (c) Repeat the calculation for $\theta = 33'$ and for $\theta = 35'$. Find the percentage change in R in each case. [4]
- (d) For small θ the approximation $\cos \theta \approx 1 - \frac{1}{2}\theta^2$ holds, with θ in radians. Use it to show that $R \approx \frac{2h}{\theta^2}$. [3]
- (e) Hence explain why an error of about 3% in θ produces an error of about 6% in R . Comment on what this means for how confidently his result can be quoted. [4]

